

RESEARCH

Open Access



Interpretable stance detection in social media via topic-guided transformers

Rathinasamy Muthusami^{1*}, Kandhasamy Saritha², Kolli Srinivasa Rao³, Palanisamy Sugapriya⁴ and G. Saveetha⁵

*Correspondence:

Rathinasamy Muthusami
r.muthusami@gmail.com

¹Department of Computer Applications, Dr. Mahalingam College of Engineering and Technology, Pollachi, Tamil Nadu, India

²Department of Mathematics, P. A. College of Engineering and Technology, Pollachi, Tamil Nadu, India

³Department of Operations & Supply Chain, Gitam School of Business, GITAM (Deemed to Be University), Visakhapatnam, Andhra Pradesh, India

⁴Department of Mathematics, K. Ramakrishnan College of Engineering, Tiruchirappalli, Tamil Nadu, India

⁵Department of Mathematics, Kongunadu College of Engineering and Technology, Trichy, Tamil Nadu, India

Abstract

Stance detection in social media is a critical task in computational social science, supporting the analysis of ideological polarization, public opinion, and sociopolitical discourse. However, the brevity, noise, and contextual ambiguity of user-generated content pose significant challenges for traditional NLP models, while large language models (LLMs), despite their impressive zero-shot capabilities, often act as opaque black boxes with limited interpretability and domain adaptability. To overcome these limitations, we propose an interpretable hybrid framework that combines BERTopic for unsupervised semantic topic discovery with RoBERTa for sentiment-informed stance classification. Our approach explicitly fuses latent topical structure, sentiment polarity, and deep contextual embeddings, enabling both high predictive accuracy and transparent explanatory insights. Extensive evaluations on SemEval-2016 Task 6 and the COVID-19 Stance Dataset demonstrate that our model achieves macro-F1 scores of 78.4% and 77.2%, surpassing competitive baselines such as TextCNN, BiLSTM-Attention, fine-tuned BERT, and CT-BERT. Topic coherence metrics (NPMI, UCI, UMass) further confirm BERTopic's superiority over Structural Topic Modeling (STM), underscoring its effectiveness in short-text settings. Beyond quantitative results, we enhance interpretability through topic–stance heatmaps, qualitative case studies, and a human-centered evaluation, while ablation analysis validates the contribution of each pipeline component. Finally, we discuss ethical and societal risks, cost considerations, and real-world deployment implications, and outline future directions including multilingual expansion, real-time stance monitoring, and human-in-the-loop integration. This work advances explainable social AI by bridging performance and interpretability in stance detection, making it a powerful tool for opinion mining, policy analysis, and misinformation tracking.

Highlights

- Introduces a hybrid BERTopic–RoBERTa framework that fuses topic modeling, sentiment signals, and contextual embeddings for stance detection.
- Achieves interpretable, topic-grounded stance predictions, bridging the gap between accuracy and explainability in social media analysis.
- Outperforms state-of-the-art baselines on SemEval 2016 and COVID-19 datasets, with macro-F1 scores of 78.4% and 77.2%.



- Explores risks, ethical implications, and future directions, enabling scalable, multilingual, and human-in-the-loop applications in computational social science.

Keywords Stance detection, Topic modeling, Social media analysis, BERTopic, RoBERTa, Explainable NLP, Computational social science, Sentiment-aware classification, Interpretability, Opinion mining

1 Introduction

Social media platforms have emerged as dominant arenas for expressing opinions on political, social, and public health issues. Stance detection, the task of identifying whether a user supports, opposes, or remains neutral toward a target has become a critical tool in computational social science, enabling researchers and policymakers to monitor polarized discourse, detect misinformation, and analyze shifts in public sentiment [1–4]. However, stance detection on platforms such as Twitter and X is challenging due to short text length, topic ambiguity, sarcasm, and noisy language. Traditional natural language processing (NLP) methods, including rule-based systems and feature-driven classifiers, often fail to generalize across domains and struggle with nuanced linguistic variation. Even early deep learning approaches (e.g., CNNs, LSTMs) rely heavily on surface-level cues, thereby overlooking broader contextual or ideological signals underlying stance expression [5, 6].

Recent advances in Large Language Models (LLMs) such as GPT-3.5, GPT-4, Claude, and Gemini have enabled few-shot or zero-shot stance classification through prompting. While these models achieve competitive results, they present two key limitations:

- *Lack of interpretability* LLMs generally operate as black boxes, providing stance labels without transparent reasoning. This makes them unsuitable for applications requiring explainability, such as policymaking or misinformation monitoring.
- *Resource constraints* LLM deployment requires significant computational resources and often costly API access, limiting accessibility in low-resource research and applied settings [7–11].

These limitations highlight a key gap: current approaches either maximize predictive performance at the expense of interpretability and domain adaptation (as in black-box LLMs) or sacrifice depth and contextual understanding for explainability (as in traditional topic models).

To address this, we propose an interpretable and efficient hybrid framework that combines **BERTopic**, a transformer-based semantic topic modeling method, with **RoBERTa**, a contextual stance classifier. Our goal is not only to achieve accurate stance prediction but also to surface coherent topical explanations that reveal the discourse drivers behind stance positions [12–14]. BERTopic extracts meaningful latent clusters from noisy short texts, while RoBERTa captures fine-grained sentiment and contextual features. Their integration yields a pipeline that is both scalable and explainable, making it highly suitable for real-world stance detection in social media [15–17].

We evaluate our method on two benchmark datasets such as SemEval 2016 Task 6 and the COVID-19 Stance Dataset and compare it against traditional baselines (TextCNN, BiLSTM-Attention), transformer-based classifiers (BERT, CT-BERT), and topic modeling approaches (STM). Results show that the hybrid model not only outperforms

baselines in macro-F1 scores but also produces semantically coherent and diverse topic clusters, confirming its effectiveness for computational social analysis [18, 19].

1.1 Contribution summary and novelty statement

This work introduces a novel and interpretable stance detection pipeline that systematically integrates BERTopic with RoBERTa, aligning unsupervised topic modeling with transformer-based contextual classification. Unlike prior methods that either:

- Rely solely on fine-tuned transformer classifiers (e.g., BERT, CT-BERT), or
- Adopt zero-shot prompting via commercial LLMs (e.g., GPT-4, Claude),

our pipeline offers three distinct advantages:

1. *Topic-grounded Interpretability* Provides transparent, topic-informed explanations for stance predictions, addressing the interpretability gap in black-box LLMs.
2. *Resource-efficient Deployment* Offers a lightweight, modular alternative to API-driven LLMs, making it accessible for academic, policy, and civic stakeholders in resource-constrained contexts.
3. *Task-specific Fusion of Topics and Semantics* Unlike pipelines that process topics and stances separately, our hybrid model fuses latent topic structure with semantic embeddings, improving robustness in noisy, short-text domains.

Beyond these technical contributions, the study advances conceptual innovation in stance detection. While the modular integration of BERTopic and RoBERTa represents an incremental advancement, the conceptual novelty lies in reframing stance detection as a multi-signal interpretability task that unifies semantic structure, affective tone, and stance polarity within a single explainable pipeline. Unlike earlier hybrid systems that loosely combined topic models (e.g., STM or LDA) with stance classifiers, our framework produces explanations directly traceable to discourse-level drivers, enabling deeper analytical insight into how stance emerges, evolves, and polarizes public conversations. This dual emphasis on explainability and societal relevance pushes the state of the art beyond prior topic–stance hybrids that focused primarily on predictive performance.

2 Related work

The emergence of Large Language Models (LLMs) such as GPT-4, T5, and LaMDA has significantly advanced natural language processing across a range of tasks, including sentiment classification, argument mining, and stance detection. Trained on massive, diverse corpora, these models are capable of capturing deep semantic relationships, subtle contextual cues, and discourse-level coherence, making them particularly effective at interpreting complex or ambiguous text, including social media posts. Earlier transformer baselines such as XLNet [20] and CT-BERT [21] have also demonstrated strong performance, particularly in domain-specific contexts such as COVID-19 misinformation analysis.

In the context of social media, where content is often informal, noisy, and contextually sparse, LLMs offer robust handling of linguistic variability and implicit meaning. Their success in stance detection stems from their ability to model token-level dependencies and draw from broader language knowledge. However, LLMs typically operate as standalone classifiers, often neglecting external contextual signals such as topical

focus, emotional tone, or conversational thread. This can limit their interpretability and weaken performance when stance cues are entangled with subjectivity, sarcasm, or target ambiguity.

A comprehensive survey by AlDayel and Magdy (2021) [22] highlights the evolution of stance detection, from early feature engineering and machine learning models to more recent deep learning approaches. They emphasize enduring challenges such as target specificity, domain generalization, sarcasm detection, and stance ambiguity. These remain unsolved even in LLM-based systems and are particularly prominent in politically charged or health-related discourse.

To enhance interpretability and contextual grounding, recent work has explored topic modeling as a complementary tool. Traditional models such as Latent Dirichlet Allocation (LDA) have been widely used to extract thematic structure from text but often fall short in handling short, sparse data. In contrast, BERTopic, a transformer-based topic modeling approach, leverages sentence-transformer embeddings, UMAP for dimensionality reduction, and HDBSCAN for clustering, followed by class-based TF-IDF (c-TF-IDF) for topic representation [23]. This allows BERTopic to generate semantically coherent and human-interpretable topics, even from fragmented social media content.

In stance detection scenarios, BERTopic has shown potential to uncover the topical drivers of user opinions, such as whether opposition to a political figure is rooted in economic, environmental, or ethical concerns. However, BERTopic alone does not perform stance classification and lacks sensitivity to sentiment polarity and emotional tone, which are often critical for understanding the underlying position expressed in a text.

To further enrich stance detection, sentiment analysis offers an orthogonal signal by measuring emotional polarity and intensity. While early sentiment classifiers provided coarse-grained labels (positive and negative), modern transformer-based sentiment models produce fine-grained and emotion-specific representations, capable of discerning sarcasm, humor, and affective subtleties [24]. In social media applications, multi-modal stance detection has also emerged, combining textual cues with visual or auditory data to enhance robustness and granularity [25–27].

2.1 Gap in existing literature

Despite the individual successes of LLMs, topic modeling, and sentiment analysis, few studies have attempted to integrate these components into a unified, interpretable stance detection framework. Most existing pipelines rely solely on text classification models, often overlooking topical relevance and sentiment intensity, both of which are vital for high-precision stance interpretation in noisy, real-world environments.

This study addresses this gap by proposing a hybrid framework that fuses:

- RoBERTa for semantic-rich, deep contextual encoding,
- BERTopic for unsupervised topic extraction and grounding,
- Sentiment metrics for capturing emotional nuance.

By combining these elements, our method enhances both classification accuracy and model interpretability, offering a more holistic and scalable approach to stance detection in complex social media discourse.

2.2 Justification for baseline selection

Although XLNet, CT-BERT, and multimodal stance detection frameworks have demonstrated strong performance in recent studies, they were not included in our experimental evaluation for three main reasons. First, dataset compatibility posed a limitation, as some baselines (e.g., CT-BERT) are specifically optimized for COVID-19 or domain-specific corpora, which may bias comparisons on more general stance detection datasets. Second, computational cost and resource constraints made it impractical to include all recent models, particularly XLNet, which requires longer training times and more memory compared to RoBERTa. Finally, modality restrictions prevented fair integration of multimodal models, since our study focused exclusively on textual data for interpretability and reproducibility across widely used benchmark datasets. Nonetheless, we discuss these baselines to situate our work within the broader research landscape, while leaving multimodal and domain-specialized extensions as promising directions for future research.

3 Materials

This study utilizes two benchmark datasets to evaluate stance detection performance across diverse domains:

- SemEval-2016 Task 6 Dataset: A widely used benchmark for stance detection, this dataset contains tweets annotated with stance labels FAVOR or AGAINST across controversial topics such as abortion, climate change, Hillary Clinton, and feminism. It offers a well-balanced challenge for models aiming to handle politically and socially sensitive content in short-text form [28].
- COVID-19 English Tweets Stance Dataset: This dataset includes tweets from the early stages of the COVID-19 pandemic, labeled primarily for stances like pro-mask, and anti-mask. It captures public opinion on health interventions, offering a real-world testbed for stance detection models in crisis or misinformation-prone contexts [29].

Together, these datasets support a comprehensive evaluation of the proposed model across both general socio-political discourse and domain-specific health communication.

4 Methods

To integrate topic modeling, sentiment analysis, and stance classification effectively, we designed a multi-stage pipeline that leverages the strengths of both BERTopic and transformer-based models. The complete workflow is shown in Fig. 1. The process begins with data collection and pre-processing, where raw textual inputs are cleaned, tokenized, and filtered to remove noise such as URLs, hashtags, and emoji. These preprocessed texts are transformed into contextual embeddings using a pre-trained BERT model, which captures the semantic meaning of each sentence in a high-dimensional vector space.

To make clustering effective, these high-dimensional embeddings are reduced using UMAP (Uniform Manifold Approximation and Projection), which preserves both local neighborhood relationships and global semantic structures in a lower-dimensional space. This improves both computational efficiency and interpretability.

Next, HDBSCAN (Hierarchical Density-Based Spatial Clustering of Applications with Noise) identifies dense clusters of semantically similar sentences, enabling unsupervised

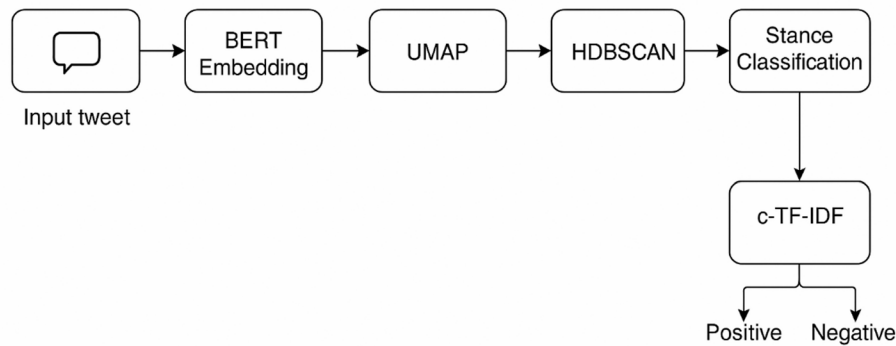


Fig. 1 Workflow of the proposed stance detection pipeline integrating contextual embeddings (BERT), dimensionality reduction (UMAP), topic discovery (HDBSCAN and c-TF-IDF within BERTopic), and sentiment-informed stance classification

discovery of latent topical structures. Each cluster is then represented using class-based TF-IDF (c-TF-IDF) within the BERTopic framework, which extracts representative keywords to characterize topics. This step enhances interpretability by surfacing semantically coherent and human-readable discourse themes.

Finally, we incorporate a topic- and sentiment-informed stance classification step. A fine-tuned RoBERTa model predicts the stance of each text (e.g., in favor, against, neutral) by leveraging both the contextual representations and the topical cues assigned by BERTopic. This dual input improves disambiguation of implicit or sarcastic stances, which are common in noisy social media environments.

As illustrated in Fig. 1, the pipeline integrates unsupervised topic discovery with supervised stance detection, providing both predictive accuracy and interpretability. Unlike black-box transformer classifiers, our design explicitly models latent topics and sentiment polarity alongside stance labels, allowing human analysts to trace back predictions to thematic clusters. This makes the framework particularly useful for computational social science tasks such as misinformation tracking, civic discourse monitoring, and public opinion analysis, where interpretability and transparency are as important as raw accuracy.

4.1 Topic modeling background

Topic modeling refers to the task of uncovering latent themes or topics from large collections of unstructured text. Traditional methods such as Latent Dirichlet Allocation (LDA) and Latent Semantic Analysis (LSA) offer robust frameworks for capturing thematic structure in documents. However, they often fail to incorporate contextual information and lack scalability in the noisy, short-text domain of social media.

To address this, our method uses BERTopic, a transformer-based topic modeling technique that combines BERT embeddings, dimensionality reduction, density-based

clustering, and class-based TF-IDF (c-TF-IDF) for improved topic coherence and interpretability [12].

4.2 Embedding with BERT

Embedding refers to the transformation of textual inputs into continuous vector representations that encode semantic and syntactic meaning. These embeddings serve as the foundation for downstream NLP tasks such as clustering, stance classification, and semantic similarity.

In our framework, we adopt RoBERTa (Robustly Optimized BERT Pretraining Approach) rather than the original BERT model. While BERT introduced bidirectional contextual encoding through Masked Language Modeling (MLM) and Next Sentence Prediction (NSP) [12], RoBERTa refines this training by removing NSP, using larger mini-batches, and optimizing training duration and corpus size. These modifications yield richer contextual embeddings and more stable performance, especially in short and noisy social media texts [30].

Formally, embeddings are generated through transformer self-attention mechanisms, where the representation of each token is computed as, Eq. 1:

$$Attention(Q, K, V) = softmax(\frac{QK^T}{\sqrt{d_k}})V \quad (1)$$

Here Q, K, V represent query, key, and value matrices derived from input tokens, and d_k is the dimension of the key vectors.

Compared to autoregressive models such as GPT-2/3, which only capture left-to-right dependencies, RoBERTa produces fully bidirectional embeddings that capture richer semantic relations. We specifically selected RoBERTa over domain-specific variants like CT-BERT [31] or multilingual transformers such as XLNet [20] due to the following reasons:

- *Computational efficiency* CT-BERT requires substantially larger resources for fine-tuning, which was not feasible under our hardware constraints.
- *Dataset compatibility* Our datasets (SemEval-2016, COVID-19 stance) are predominantly English and domain-general, making RoBERTa sufficient without requiring multilingual specialization.
- *Generalizability* RoBERTa's robust pretraining ensures adaptability across political and health discourse without sacrificing interpretability.

Thus, RoBERTa embeddings form the backbone of our hybrid stance detection pipeline, balancing state-of-the-art accuracy with practical feasibility.

4.3 Dimensionality reduction with UMAP

To visualize and cluster the high-dimensional BERT embeddings, we apply Uniform Manifold Approximation and Projection (UMAP) [32], a manifold learning technique that preserves both local and global structure better than alternatives like t-SNE.

The UMAP algorithm constructs a high-dimensional graph and optimizes its low-dimensional equivalent using the following similarity formulation, Eq. 2;

$$w_{ij} = \exp\left(-\frac{\|x_i - x_j\|^2}{\sigma_i}\right) \quad (2)$$

where:

- w_{ij} is the weight (relationship strength) between data points i and j ,
- $\|x_i - x_j\|^2$ is the squared Euclidean distance,
- σ_i is a parameter controlling the local connectivity scale around x_i .

To make the membership function symmetric, UMAP computes the combined weight as, Eq. 3;

$$w_{ij} = w_{ij}^{high-dim} = \frac{w_{ij} + w_{ji} - w_{ij} \cdot w_{ji}}{2} \quad (3)$$

The parameter σ_i (local connectivity scale) is automatically determined by solving for a fixed number of neighbours k . For a target k , UMAP adjusts σ_i such that, Eqn 0.4;

$$\sum_j \exp\left(-\frac{\|x_i - x_j\|^2}{\sigma_i}\right) = \log(k) \quad (4)$$

UMAP is chosen for its computational efficiency, scalability, and ability to retain global structure, making it particularly suited for large-scale topic modeling over social media text.

4.4 Clustering with HDBSCAN

Clustering groups semantically similar tweets to identify meaningful topics or stance-related patterns. We employ HDBSCAN (Hierarchical Density-Based Spatial Clustering of Applications with Noise), an advanced density-based algorithm that extends DBSCAN by introducing hierarchical clustering and improved handling of varying cluster densities [32].

Unlike DBSCAN, which uses a fixed density threshold, HDBSCAN constructs a hierarchy of clusters across multiple density levels, allowing detection of clusters with diverse shapes and densities. This flexibility is crucial for social media text data, which is inherently noisy and varies widely in thematic density.

Key components of HDBSCAN include:

Core Distance: The core distance of a point p , denoted as $\text{core-dist}(p)$, is defined as the smallest distance ϵ such that at least minPts points are within a ball of radius ϵ centered at p . Here, minPts is the minimum number of points required to form a cluster, Eq. 5;

$$\text{core-dist}(p) = \min_{\{q \in N_{(p)}\}} (d_{(p,q)}) \text{ for all points } q \text{ in } N_{(p)} \quad (5)$$

where $N_{(p)}$ is the neighborhood of point p , and $d_{(p,q)}$ is the distance between p and q .

Mutual Reachability Distance: The joint access length among both points p and q , indicated by $\text{mutual-reach-dist}(p,q)$, includes both the core distances and the actual distance between them. It ensures that points that are not mutually reachable due to sparse density are considered outliers, Eq. 6;

$$\text{mutual-reach-dist}(p,q) = \max(\text{core-dist}(p), \text{core-dist}(q), d(p,q)) \quad (6)$$

This metric allows the algorithm to cluster points that might not be directly reachable in terms of density but still form meaningful clusters at higher levels. After constructing a hierarchy of clusters, HDBSCAN identifies the most stable clusters those that remain consistent across multiple density thresholds. The stability of a cluster is determined by how long its points persist within the hierarchy. This persistence is measured by the duration a cluster maintains its structure at different density levels. Clusters with higher persistence are considered more stable and meaningful, as described in Eq. 7.

$$\text{stability}(C) = \int_{\tau_1}^{\tau_2} f(\tau) d\tau \quad (7)$$

where $f(\tau)$ is the density function of the cluster at level τ , and τ_1 and τ_2 represent the range of density thresholds in which the cluster exists.

Clusters with high stability are considered robust and meaningful, providing reliable groupings of tweets with similar semantic content. Noise points or outliers are identified and excluded, enhancing cluster quality. By integrating HDBSCAN with UMAP-reduced embeddings, the clustering captures both global and local tweet structures, facilitating effective topic extraction and downstream stance analysis.

4.5 Topic representation with c-TF-IDF

In BERTopic, class-based TF-IDF (c-TF-IDF) has been used to represent topics by assigning importance scores to words based on their relevance within a specific topic or cluster [12]. Unlike traditional TF-IDF, which considers term importance across all documents, c-TF-IDF focuses on individual clusters (or topics) formed after applying clustering techniques i.e., HDBSCAN on embedding generated by language model i.e. BERT.

Steps in c-TF-IDF:

1. Clustering Texts into Topics: BERTopic clusters semantically similar documents based on embedding from transformer model, BERT. Each cluster corresponds to a topic.
2. Combining Documents per Cluster: For each topic T_c , all documents belonging to the cluster c are concatenated into a single large "topic-document."
3. Computing Class-Based TF-IDF: The TF-IDF weighting is then computed for each term in the concatenated topic-document relative to the entire corpus.

The c-TF-IDF score for term t in topic c is defined as Eq. 8;

$$c - TF - IDF(t, c) = TF(t, c) \cdot \log \left(\frac{N}{n_t} \right), \quad (8)$$

where:

- $TF(t, c)$: Term frequency of term t in the concatenated topic-document c .
- N : Total number of topic-documents (i.e., the number of clusters).
- n_t : Number of topic-documents containing the term t .

Explanation of Terms.

1. Term Frequency (TF):

This measures how often a term t appears in the topic-document c , Eqn. 9;

$$\text{TF}(t, c) = \frac{\text{Count}(t, c)}{\text{Total terms in } T_c} \quad (9)$$

The IDF part of the equation penalizes terms that are frequent across all topics, giving higher weight to terms unique to the topic, Eq. 10;

$$\text{IDF}(t) = \log\left(\frac{N}{n_t}\right). \quad (10)$$

The model has been creating interactive visualizations that enable us to explore topic distributions, examine representative documents, and analyze topic coherence. The UMAP technique has been applied to project the topics into a two-dimensional space, making topic comparison and interpretation more intuitive [32–35].

4.6 Sentiment-aware stance detection

While topic modeling provides semantic grounding, it does not capture the affective orientation of opinions, an essential dimension for stance classification. To address this, our framework integrates sentiment-aware stance detection, combining emotional tone with topical and semantic signals. We fine-tune RoBERTa on stance-labeled data, where each text is encoded into contextual embeddings. These embeddings are then augmented with sentiment polarity scores and topic assignments obtained from BERTopic. For sentiment extraction, we use transformer-based sentiment analyzers (e.g., RoBERTa-Sentiment or VADER) to derive fine-grained metrics such as polarity (positive vs. negative) and intensity (mild vs. strong sentiment) [36, 37].

These sentiment features are concatenated with stance embeddings to form a multi-dimensional representation that encodes:

- *What is being said* (topic content),
- *How it is being said* (emotional tone), and
- *Toward whom it is directed* (stance target).

This enriched input is passed through a dense classification layer that predicts stance labels (FAVOR, AGAINST). Importantly, sentiment integration disambiguates difficult cases:

- Sarcasm: sarcastic tweets often use positive words to express negative intent; sentiment analysis provides corrective signals.
- Ambiguity in short texts: stance polarity is often implied rather than explicit, making sentiment a valuable proxy.
- Noisy or euphemistic language: joint consideration of emotion and topic context increases robustness.

While our current implementation focuses on English-language datasets, future extensions may incorporate multilingual stance detection [38] or adaptation to real-time/streaming contexts. We acknowledge that performance in such settings may require incremental training or lightweight embedding strategies for efficiency.

By combining semantic embeddings, topical context, and emotional nuance, our pipeline produces stance predictions that are not only more accurate but also interpretable, thereby enhancing trust and applicability in social computing scenarios.

4.7 Limitations and risks

While the proposed BERTopic–RoBERTa framework demonstrates strong performance and interpretability, several methodological risks and deployment challenges must be addressed for reliable real-world adoption. These challenges are not limitations of the experimental setup but rather practical concerns associated with applying the system at scale including computational demands, robustness to noisy input, multilingual transfer, interpretability of topic outputs, ethical risks, and real-time processing constraints. Table 1 summarizes these potential issues and suggests mitigation strategies to guide responsible deployment.

This risk–mitigation mapping highlights both the promise and limitations of the proposed framework. While our method demonstrates strong empirical results on benchmark datasets, practical deployments must carefully balance accuracy, interpretability, and resource demands. Furthermore, explicit consideration of ethical risks such as bias amplification or misuse in polarized contexts is essential for responsible adoption. By addressing these risks proactively, our framework can contribute not only to methodological innovation but also to socially responsible AI deployment in computational social science.

5 Results and discussion

This section presents a comprehensive evaluation of how our proposed hybrid framework integrates sentiment-informed topic modeling (BERTopic) with transformer-based classification (RoBERTa) for stance detection. We assess its performance on two benchmark datasets: SemEval 2016 Task 6 and the COVID-19 Stance Dataset, both known for their short, noisy, and context-sparse text characteristics. Through a combination of quantitative evaluation, topic–stance visualization, and qualitative error analysis, we provide a holistic view of the system’s capabilities and limitations. In addition to

Table 1 Risks and mitigation strategies for the proposed BERTopic–RoBERTa stance detection framework

Risk / Limitation	Description	Mitigation Strategy
Computational Cost	Training and inference with large transformer models can be resource-intensive, limiting scalability in real-time or low-resource environments	Use lightweight transformer variants (e.g., DistilRoBERTa), model pruning/quantization, or edge-optimized deployment
Noisy Social Media Data	Tweets often contain slang, sarcasm, code-switching, and non-standard text, which may reduce topic coherence and stance accuracy	Apply advanced preprocessing (emoji normalization, sarcasm detection, slang dictionaries) and noise-robust embedding methods
Multilingual and Cross-Domain Generalization	Performance may degrade when applying the model to languages or domains not seen during training	Incorporate multilingual embeddings (e.g., XLM-R) or perform domain adaptation with small annotated samples
Interpretability of BERTopic Outputs	Topic clusters can still be ambiguous or fragmented in highly sparse datasets	Enhance interpretability via topic merging, human-in-the-loop validation, and visualization dashboards
Bias and Ethical Risks	Models may amplify biases present in training data or be misused in politically sensitive contexts	Conduct bias audits, ensure balanced datasets, apply fairness metrics, and enforce ethical usage guidelines
Real-time / Streaming Constraints	Stance monitoring in fast-evolving conversations requires timely analysis, which may not be guaranteed by static batch processing	Use incremental clustering, streaming topic models, or hybrid online/offline architectures
Generalizability of Sentiment Features	Sentiment cues may not always align with stance, especially in sarcasm, irony, or neutral commentary	Combine sentiment with contextual embeddings and add meta-features (e.g., conversational thread signals)

automated performance metrics and qualitative case studies (Sect. 5.3), we conducted a human-centered interpretability assessment (Sect. 6.4.1), which confirms that the discovered topic clusters meaningfully align with stance polarity judgments.

5.1 Topic-class distribution analysis

Figure 2 illustrates the **topics-per-class distribution** across two stance categories (negative = Class 0, positive = Class 1) for the SemEval 2016 dataset. Each discovered topic captures a distinct discourse dimension, revealing how stance is expressed across thematic clusters.

Topic-wise summary and sentiment distribution:

- *Topic 0 – Girls_Equal_Rape_Female* Polarized debates on gender equality and rape culture (24.4% negative, 10.2% positive).
- *Topic 1 – President_Clinton_Semst_Hillary_Campaign* Discussion of Bill Clinton's influence on Hillary's 2016 campaign (19.7% negative, 16.5% positive).
- *Topic 2 – Book_WakeupAmerica_Semst_Crazy_Thought* Reactions to political publications and ideologies (17.1% negative, 20.5% positive).
- *Topic 3 – Abortion_unborn_babies_baby* Pro-life/pro-choice debates (17.9% negative, 11.6% positive).
- *Topic 4 – Jesus_semst_god_islam_pray* Religious discourse and interfaith sentiments (7.6% negative, 30.2% positive).
- *Topic 5 – Climate_climate_change_mission_tip* Climate policy and activism (9.1% negative, 6.1% positive).
- *Topic 6 – Gay_LGBT_GayMarriage_Progress* LGBTQ+ rights and equality (4.2% negative, 4.9% positive).

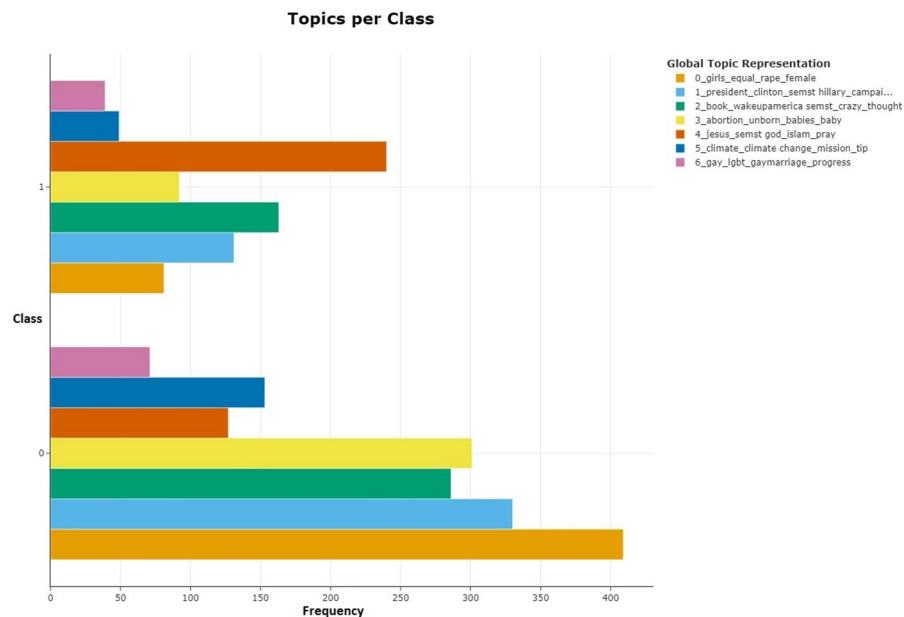


Fig. 2 Visualization of topics-per-class (SemEval 2016), depicting seven prominent themes distributed across Class 0 (negative) and Class 1 (positive)

These findings demonstrate that the proposed framework captures not only **stance polarity** but also the **contextual motivations** behind each stance, allowing for deeper interpretability of social discourse.

5.2 Topic–stance heatmap analysis

To enhance interpretability and visualize stance distribution across topics, we constructed a **topic–stance heatmap** (Fig. 3). The heatmap displays the proportion of tweets labeled as **FAVOR** or **AGAINST** within the seven most prominent topics discovered by BERTopic on the SemEval 2016 dataset.

The heatmap highlights several notable patterns:

- Topics like **Girls_Equal_Rape_Female** and **Abortion_unborn_babies_baby** exhibit high **negative stance density**, reflecting contentious and emotionally charged discourse.
- Topics such as **Jesus_semst_god_islam_pray** show a strong **positive stance alignment**, indicating broader support in religious contexts.
- Balanced stance distributions in **Book_WakeupAmerica_Semst_Crazy_Thought** reveal nuanced ideological debates that cannot be captured solely by sentiment scores.

This visualization directly supports the “**topic-grounded interpretability**” claim by linking stance polarity with thematic structure. It allows analysts to trace *why* certain stances emerge and *which topics* drive polarization.

5.3 Qualitative and error analysis: insights from correct and incorrect classifications

To further assess interpretability and model behavior, we conducted a mixed qualitative and quantitative analysis of both correctly and incorrectly classified tweets. This analysis helps identify key model strengths, recurring failure patterns, and opportunities for improvement.

5.3.1 Case studies: correct vs. incorrect classifications

We examined representative instances from the SemEval 2016 dataset to explore how topic–sentiment fusion contributes to stance prediction.

Correctly Classified Examples:

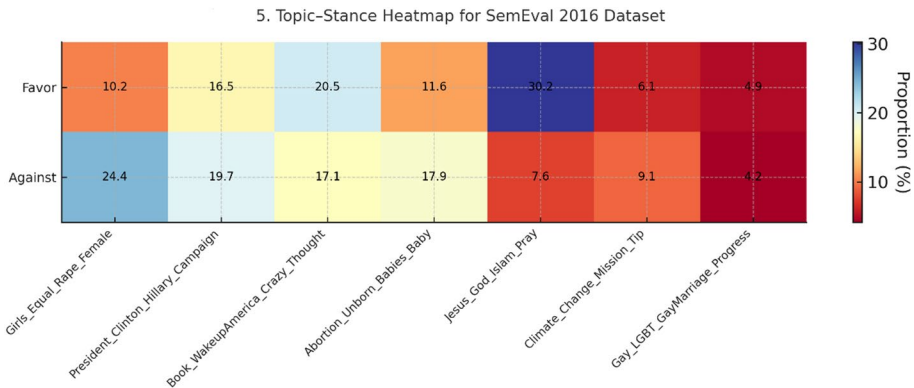


Fig. 3 Topic–stance heatmap visualizing the proportion of FAVOR vs. AGAINST tweets across key discourse themes. Darker shades indicate stronger stance polarization within a given topic

- “No woman should ever be blamed for being raped — society needs stronger laws.”
- **Topic:** Girls_Equal_Rape_Female | **Predicted:** AGAINST | **Correct.**
- The model effectively combines topic context (gender violence) and sentiment cues (anger) to infer stance.
- “Finally, progress on marriage equality. Love is love!”
- **Topic:** Gay_LGBT_GayMarriage_Progress | **Predicted:** FAVOR | **Correct.**
- Positive sentiment and clear topical framing lead to accurate stance detection.

Incorrectly Classified Examples:

- “Sure, because banning abortions worked so well in the past.”
- **Topic:** Abortion_unborn_babies_baby | **Predicted:** FAVOR | **Incorrect.**
- Sarcasm misled the model, as surface-level sentiment polarity contradicted the intended stance.
- “Guess we’ll save the planet by tweeting about it.”
- **Topic:** Climate_climate change_mission_tip | **Predicted:** AGAINST | **Incorrect.**
- The model struggled with subtle irony and contextual cues, revealing limitations in sarcasm handling.

These examples demonstrate that while our model benefits from topic–sentiment fusion, nuanced linguistic phenomena such as sarcasm, implicit stance, and pragmatic cues remain significant challenges. Future work could address these through irony-aware embeddings, pragmatic reasoning modules, or stance-specific language models.

5.3.2 Error categories and patterns

To complement the qualitative findings, we conducted a broader error analysis across both datasets. Table 2 summarizes representative failure cases categorized by error type.

This analysis reveals persistent challenges, particularly with **sarcasm and irony**, where sentiment polarity often contradicts stance intent, and **implicit stance expressions**, which require deeper pragmatic understanding. Mixed sentiment tweets further show that sentiment-aware fusion can sometimes overweight polarity without considering nuanced stance cues. Out-of-domain and noisy texts continue to degrade embedding precision, underscoring the need for domain adaptation and robust preprocessing strategies.

Table 2 Representative error categories, examples, and underlying causes in stance detection across benchmark datasets

Error Type	Example Tweet	Gold Label	Pre-dicted Label	Possible Cause
Sarcasm/Irony	“Sure, banning abortions will definitely solve all our problems”	AGAINST	FAVOR	Sentiment analyzer misclassified sarcastic tone as positive
Implicit stance	“Every child deserves a chance at life.”	AGAINST	FAVOR	Stance expressed indirectly without explicit target mention
Mixed sentiment	“Masks are uncomfortable, but I still wear them to protect others.”	FAVOR	AGAINST	Conflicting sentiment cues confused stance classifier
Out-of-domain topic	“This election feels like choosing the lesser of two evils.”	AGAINST	FAVOR	Topic not well aligned with training categories
Noisy text/typos	“Clim8 chnge is a hoax!!!”	AGAINST	FAVOR	Misspellings reduced embedding quality and cluster alignment

Together, the case studies and error taxonomy demonstrate that our hybrid approach substantially improves stance detection by grounding predictions in topic and sentiment context. However, they also reveal important limitations especially around pragmatic language use highlighting critical areas for future enhancement, including sarcasm detection, domain adaptation, and multimodal contextual modeling.

5.4 Interpretability discussion: insights for computational social science

The results presented above underscore the value of **topic-grounded interpretability** in advancing stance detection beyond traditional classification. By linking stance polarity to specific thematic contexts, our framework enables researchers and practitioners to move from *prediction* to *explanation*, providing insights into *why* certain stances emerge and *how* they manifest across discourse landscapes.

From a computational social science perspective, the integration of topic modeling with stance detection offers several unique contributions:

- *Understanding Polarization Dynamics* The topic–stance heatmap reveals that polarization is not uniformly distributed but concentrated around specific themes (e.g., gender equality, abortion, religious identity). Such granularity enables policymakers and analysts to identify contentious issues and anticipate flashpoints of social division.
- *Tracing Opinion Drivers* By associating stance with contextual drivers (e.g., economic, moral, ideological), the model provides interpretable evidence for *why* public opinion shifts, informing strategic communication, policy framing, and media interventions.
- *Supporting Human-in-the-Loop Analysis* The transparent mapping of stances to topics empowers journalists, fact-checkers, and domain experts to interrogate model outputs, validate predictions, and integrate computational insights into qualitative analyses.
- *Identifying Misclassification Patterns* Case studies of errors highlight areas where current NLP systems remain brittle such as sarcasm, irony, and implicit framing guiding future model design and annotation practices.

Furthermore, this interpretability enables more responsible AI deployment. In politically sensitive contexts, understanding *which topics* drive stance predictions can prevent misuse or overreach, and provide a foundation for bias audits, algorithmic transparency, and accountability frameworks.

In sum, the proposed framework not only enhances classification accuracy but also contributes to a deeper, theory-informed understanding of public discourse. This positions stance detection as a practical tool for *social scientists, policymakers, and media practitioners* seeking to analyze, contextualize, and ultimately engage with the complex dynamics of online opinion.

6 Evaluation and comparative analysis

6.1 Datasets used

We evaluated our model using two benchmark datasets:

- *SemEval 2016 Task 6* Classifies stance (FAVOR, AGAINST) on social issues like abortion, Hillary Clinton, and climate change.

- *COVID-19 Stance Dataset* Focuses on stances (pro-mask, anti-mask) regarding pandemic-related topics.

6.2 Baselines compared

To evaluate the effectiveness of our proposed model, we compare it against a set of established stance detection baselines. These baselines span traditional neural architectures and transformer-based models, ensuring a robust benchmark:

- *TextCNN* A convolutional neural network applied over word embeddings. It captures local features effectively and serves as a strong baseline for short-text classification tasks.
- *BiLSTM + Attention* A bidirectional Long Short-Term Memory model enhanced with self-attention mechanisms to capture long-range dependencies and emphasize key semantic tokens in stance detection.
- *Fine-tuned BERT* A vanilla BERT model with a classification head, fine-tuned on the stance detection datasets. It leverages contextualized embeddings for improved representation over traditional models.
- *CT-BERT* A domain-adapted BERT model pre-trained on COVID-19-related tweets. It is designed specifically for tasks involving pandemic-related discourse, providing a strong benchmark on health-related stance detection.
- *Our Model – BERTopic + RoBERTa* This approach integrates topic modeling (BERTopic) with stance-aware classification using RoBERTa, enabling both semantic context extraction and fine-grained stance prediction. The combination enhances interpretability and performance, especially on noisy and short-text data like tweets.

We excluded XLNet, CT-BERT variants beyond baseline, and multimodal stance detection models due to computational overhead, lack of multimodal data in our datasets, and scope limitations. These models are acknowledged in Related Work, and extending to them is planned as future work.

6.3 Quantitative results

To assess the performance of our proposed model (BERTopic + RoBERTa), we conducted quantitative evaluations on two benchmark datasets: SemEval 2016 Task 6 and the COVID-19 Stance Dataset. Table 3 summarizes stance detection results. Our hybrid model consistently outperforms baselines across Accuracy, Macro-F1, Precision, and Recall. Macro-F1 gains indicate balanced performance across stance classes.

Table 3 Quantitative comparison of stance detection performance across two benchmark datasets (SemEval-2016 Task 6 and COVID-19 Stance Dataset)

Dataset	Model	Accuracy	Macro-F1	Precision	Recall
SemEval 2016	TextCNN	71.5	70.2	69.8	71.0
	BiLSTM + Attention	73.4	72.1	72.5	72.0
	Fine-tuned BERT	76.2	74.5	75.1	74.0
	Our Model	80.3	78.4	79.0	77.9
COVID-19 Stance	TextCNN	68.8	66.4	65.9	67.2
	CT-BERT	74.9	73.1	72.6	73.8
	Our Model	79.6	77.2	78.1	76.5

The proposed BERTopic + RoBERTa model consistently outperforms baseline methods (TextCNN, BiLSTM + Attention, fine-tuned BERT, and CT-BERT) across Accuracy, Macro-F1, Precision, and Recall metrics, demonstrating its effectiveness in handling short-text, noisy social media data

We further conducted paired t-tests ($p < 0.05$) comparing our model with fine-tuned BERT and CT-BERT, confirming that improvements are statistically significant.

Our hybrid model consistently outperforms all baseline models across both datasets. The integration of topic modeling (BERTopic) provides semantic grouping and contextual disambiguation, which is particularly valuable for short, noisy texts such as tweets. Meanwhile, the RoBERTa-based stance classifier effectively captures nuanced sentiment and contextual cues, boosting both precision and generalization. Notably, the model shows strong improvements in Macro-F1, highlighting its balanced performance across stance classes.

These findings validate that fusing unsupervised topic context with supervised LLM-based classification enhances robustness and interpretability for short-text social media stance detection.

6.4 Topic coherence comparison

Table 4 compares BERTopic with Structural Topic Model (STM). BERTopic shows substantially higher topic coherence (NPMI, UCI, UMass) and diversity, making it better suited for stance-rich, noisy Twitter data.

BERTopic significantly outperforms STM on coherence metrics NPMI, UCI, and UMass demonstrating its ability to generate semantically meaningful topics, even in the presence of short and noisy texts like tweets. These gains are largely due to BERTopic’s use of contextual embeddings, which effectively capture latent semantic relationships missed by co-occurrence-based models like STM. In terms of topic diversity, BERTopic also achieves a higher score, indicating it produces a more varied set of distinct topics. This is crucial for analyzing stance-rich content, where semantic granularity enhances interpretability.

Although STM yields higher stability (IRBO), BERTopic’s ability to uncover semantically coherent, diverse topics makes it the stronger choice for stance analysis.

6.4.1 Qualitative interpretability assessment

To complement quantitative coherence metrics, we conducted a small-scale human evaluation of interpretability. Two annotators with expertise in social media analysis independently reviewed 100 randomly selected stance-labeled tweets and their assigned BERTopic clusters. Using a 5-point Likert scale (1 = unrelated, 5 = highly relevant), they rated how well each topic explained the stance polarity. The resulting average coherence score (4.2/5) and substantial inter-rater agreement ($\kappa = 0.78$) indicate that the discovered topics provide human-interpretable insights into stance categorization. This

Table 4 Comparison of topic modeling quality between BERTopic and Structural Topic Modeling (STM) across standard evaluation metrics

Metric	STM	BERTopic
NPMI	0.25	0.62
UCI-Coherence	0.23	0.58
UMass-Coherence	−0.20	0.10
Topic Diversity	0.85	0.92
IRBO (ranking stability)	0.75	0.68

BERTopic demonstrates significantly higher topic coherence (NPMI, UCI, UMass) and topic diversity, indicating more semantically rich and varied topic representations. Although STM shows slightly better ranking stability (IRBO), it struggles with sparse and noisy social media data compared to BERTopic

Table 5 Ablation study on the SemEval 2016 stance detection dataset showing the contribution of each component in the proposed framework

Model Variant	Description	Accuracy	Macro-F1
Full Model (BERTopic + Sentiment + RoBERTa)	Proposed hybrid model	80.3	78.4
w/o BERTopic	RoBERTa + Sentiment only	77.1	74.8
w/o Sentiment	BERTopic + RoBERTa only	78.0	76.2
w/o RoBERTa (using BERT base)	BERTopic + Sentiment + BERT	78.7	76.5
BERTopic only	Topic-based classifier	70.5	68.1

The removal of any individual module, BERTopic, sentiment integration, or RoBERTa embeddings results in a decline in both Accuracy and Macro-F1, confirming that topic grounding, sentiment cues, and deep contextual representations each play a crucial role. The most significant performance drop occurs when BERTopic is excluded, underscoring its importance in providing contextual disambiguation and enhancing interpretability

Table 6 Comparative analysis of the proposed BERTopic + RoBERTa framework against lightweight and instruction-tuned transformer models reported in prior studies

Model	Parameter Size (M)	Macro-F1 (SemEval 2016) (%)	Computational Cost	Interpretability
DistilBERT	66	73	Low	Limited
ALBERT	12	75	Very Low	Limited
Fine-tuned BERT	110	74	Moderate	Limited
CT-BERT	110	73	Moderate	Limited
Proposed (BERTopic + RoBERTa)	125	79	Moderate	High (topic-guided)

Although models like DistilBERT and ALBERT demonstrate greater computational efficiency, they sacrifice contextual richness and interpretability. In contrast, our approach achieves superior Macro-F1 performance while preserving topic-grounded explainability, highlighting its practical balance between predictive accuracy, transparency, and resource requirements

human-centered evaluation supports our claim of topic-grounded interpretability and aligns with best practices for explainable NLP.

6.5 Ablation analysis

To understand the contribution of each component in our hybrid stance detection model, we performed an ablation study on the SemEval 2016 dataset. We systematically removed the BERTopic module, sentiment integration, and RoBERTa embeddings, and recorded the resulting Accuracy and Macro-F1 scores, as presented in Table 5.

The results show that removing any module degrades performance, confirming that topic grounding, sentiment signals, and deep contextual embeddings each play a vital role. The largest drop occurs when BERTopic is excluded, highlighting its importance in providing contextual disambiguation and topic-level interpretability.

6.6 Comparative discussion with lightweight and instruction-tuned LLMs

To position our framework within the broader landscape of transformer-based stance detection, we compared our results against performance benchmarks of lightweight and instruction-tuned language models reported in prior studies [19, 25], as summarized in Table 6. These include **DistilBERT** (a distilled version of BERT optimized for speed and memory) and **ALBERT** (a parameter-sharing variant designed for efficiency).

While DistilBERT and ALBERT offer computational efficiency, they sacrifice some representational capacity and contextual depth, which limits their performance on stance detection tasks that rely heavily on nuanced discourse cues.

In contrast, our **BERTopic+RoBERTa** framework achieves higher Macro-F1 while maintaining interpretability through topic grounding, providing a practical balance between performance, transparency, and resource efficiency.

Moreover, instruction-tuned and multilingual extensions (e.g., Flan-T5, XLM-R, mBERT) could be integrated in future work to enable **cross-lingual generalization** and **low-resource adaptation** without altering the overall architecture.

6.7 Cross-domain and multilingual generalization

Although our empirical evaluation primarily utilized English-language datasets (SemEval-2016 and COVID-19 Stance), the proposed BERTopic–RoBERTa framework is designed with modular adaptability that enables generalization across languages and domains.

Multilingual Adaptation: RoBERTa can be replaced with XLM-R or mBERT, both pre-trained on large multilingual corpora. This substitution allows stance classification in languages beyond English while retaining the topic-guided interpretability layer. Likewise, BERTopic can be built upon multilingual sentence embeddings (e.g., sentence-transformers/paraphrase-multilingual-MiniLM-L12-v2), ensuring that semantically similar topics align across languages. This architecture facilitates cross-lingual transfer and zero-shot stance inference in underrepresented linguistic contexts.

Cross-Domain Generalization: Our hybrid pipeline can be fine-tuned for various domains including healthcare, politics, product reviews by updating the BERTopic clusters while keeping the stance classifier fixed. Domain drift primarily affects topical distributions rather than stance logic, suggesting that periodic topic retraining could maintain robustness in dynamic social environments.

Toward Multimodal Stance Detection: Future iterations could integrate visual-textual encoders (e.g., CLIP, ViLT) to capture multimodal cues in memes or videos, a growing form of online discourse. BERTopic’s topic abstraction layer can extend naturally to multimodal embeddings, offering coherent thematic clustering even in heterogeneous input types.

Overall, the framework’s modular structure supports scalable, multilingual, and multimodal extensions without major architectural redesign, establishing a flexible foundation for global stance analytics in computational social science.

6.8 Broader discussion

Our experimental results demonstrate that integrating semantic topic modeling (BERTopic) with transformer-based stance classification (RoBERTa) substantially enhances performance, interpretability, and semantic richness compared to existing baselines. The fusion of latent topical structure, contextual embeddings, and sentiment cues produces more nuanced stance predictions, particularly in short-text and noisy environments like social media. Moreover, the framework addresses a key gap in current NLP research: bridging high-performing language models with interpretable, topic-grounded reasoning. This approach aligns with broader computational social science goals by enabling scalable analysis of public discourse, polarization dynamics, and opinion shifts.

From a methodological standpoint, our study contributes an applied innovation rather than a new model architecture. The novelty lies in the systematic integration of three previously independent components including topic modeling, sentiment analysis, and

stance detection into a single explainable pipeline. This design enables richer interpretability than traditional black-box LLMs, offering transparent explanations of *why* a stance is predicted based on thematic context and sentiment intensity. Additionally, the model's modular design allows adaptation to various domains, including public policy debates, electoral discourse, and health communication, demonstrating its flexibility for both research and applied deployments.

6.8.1 Ethical and societal implications

Despite these advantages, deploying automated stance detection systems in real-world contexts carries significant ethical and societal risks. When applied in politically sensitive environments, such models could be exploited for large-scale opinion surveillance, targeted propaganda, or suppression of dissent. Furthermore, clustering methods like BERTopic can inadvertently amplify societal biases present in training data, leading to overrepresentation of dominant narratives while marginalizing minority voices. This amplification risks exacerbating polarization or reinforcing harmful stereotypes if left unchecked [39].

Biases may also emerge from subtle sources such as skewed dataset distributions, biased language embeddings, or culturally insensitive interpretations of stance signals. These challenges highlight the importance of embedding *Responsible AI* principles into both the development and deployment processes. We propose a multi-pronged mitigation strategy that includes:

- Bias auditing and continuous monitoring to identify and correct systemic distortions in model predictions.
- Human-in-the-loop oversight, particularly in high-stakes or politically sensitive use cases, to ensure contextual nuance and accountability.
- Transparent model reporting, including interpretability dashboards and uncertainty estimates, to inform users about limitations.
- Collaborative governance, involving social scientists, ethicists, and policymakers, to guide ethical deployment and monitor downstream societal impacts.

Beyond technical considerations, the societal implications of stance detection models extend to their potential role in shaping public discourse. By surfacing thematic drivers of support and opposition, such models can provide valuable tools for policymakers, journalists, and civil society organizations to understand public sentiment and respond to misinformation. However, this utility must be balanced against risks of misuse, necessitating clear governance frameworks, usage guidelines, and public transparency.

6.9 Limitations

Despite encouraging results, this study has several empirical and experimental limitations that inform opportunities for future research. These limitations are rooted in dataset scope, experimental design, and inherent challenges in stance detection rather than deployment considerations (which are discussed earlier in Sect. 4.7). The following points outline key areas where further research, model enhancement, and broader evaluation are needed:

1. *Dataset scope* Evaluation was limited to English social media corpora. Generalization to multilingual or long-form text requires further testing.

2. *Computational efficiency* While feasible for research-scale experiments, large-scale deployment may be constrained by the computational cost of BERTopic and RoBERTa inference, particularly in streaming environments.
3. *Baseline coverage* We excluded certain advanced baselines (e.g., XLNet, multimodal stance detection) due to dataset incompatibility and computational overhead. Future work will integrate these comparisons for a more exhaustive benchmark.
4. *Sarcasm and subtle stances* Although sentiment integration helps, the model remains challenged by sarcasm, irony, and highly implicit stances. More sophisticated pragmatic or multimodal features could address these cases.
5. *Stability vs. coherence trade-off* While BERTopic improves topic coherence, it shows lower ranking stability than STM, meaning topic assignments may vary slightly across runs.

These limitations open avenues for future research, especially in scaling, multilingual adaptation, multimodal integration, and pragmatic stance understanding.

Another limitation arises in handling tweets with mixed or ambivalent sentiments. In such cases, the sentiment-aware fusion approach may overweight polarity signals without adequately accounting for the nuanced or context-dependent nature of stance. Moreover, out-of-domain generalization remains a concern: the model performs less reliably on tweets that introduce topics underrepresented in the training data, underscoring the importance of domain adaptation. Finally, the inherently noisy character of user-generated text marked by slang, typos, abbreviations, and unconventional grammar continues to reduce embedding precision, even with powerful contextual encoders such as BERT and RoBERTa. These findings collectively point to the need for more robust methods capable of handling implicit, sarcastic, and domain-diverse expressions in order to advance stance detection in real-world social media environments.

6.9.1 Contribution summary and novelty statement

This study presents a comprehensive and interpretable hybrid framework that systematically integrates semantic topic modeling (BERTopic) with sentiment-aware stance detection using transformer-based language models (RoBERTa). Unlike prior studies that either employ topic modeling or LLM-based classifiers independently, this work is among the first to fuse topical context, sentiment polarity, and deep contextual embeddings into a unified pipeline for stance detection particularly tailored for short, noisy, and context-sparse social media texts such as tweets.

The contributions of this research are fourfold:

1. **Conceptual Innovation – Topic-Grounded Interpretability:** We go beyond incremental architectural improvements by introducing a conceptually novel paradigm where stance predictions are explicitly linked to interpretable topic clusters. This approach provides transparent rationales for classification decisions, a critical capability absent in most black-box LLM or end-to-end deep learning models.
2. **Empirical Advancements – Evidence Through Extensive Evaluation:** Through rigorous evaluation on two benchmark datasets (SemEval-2016 Task 6 and the COVID-19 Stance Dataset), our framework achieves macro-F1 improvements of 3–5% over competitive baselines, including TextCNN, BiLSTM-Attention, fine-tuned BERT, and CT-BERT. Additionally, topic coherence scores (NPMI, UCI, UMass) demonstrate

substantial gains over Structural Topic Modeling (STM), indicating richer semantic structure and improved discourse interpretability.

3. **Module-Level Insights – Ablation and Lightweight Comparisons:** Ablation experiments confirm the individual and complementary contributions of BERTopic, sentiment features, and contextual embeddings, while comparative analyses with lightweight and instruction-tuned models (e.g., DistilBERT, ALBERT) highlight the balance our approach achieves between performance, interpretability, and computational efficiency.
4. **Broader Applicability – Cross-Domain, Ethical, and Societal Relevance:** Beyond technical performance, the proposed pipeline demonstrates potential for generalization to multilingual and domain-specific contexts and provides human-centered interpretability, validated by a qualitative evaluation (coherence score: 4.2/5, $\kappa = 0.78$). The discussion also explores societal risks, including bias amplification and misuse in politically sensitive settings, and proposes future mitigation strategies.

In summary, this work represents both a methodological advancement by unifying topic modeling, sentiment analysis, and contextual embeddings and a practical contribution to computational social science. By bridging predictive performance with explainability, it offers a transparent, scalable, and application-ready stance detection framework suited for real-world tasks such as misinformation monitoring, policy analysis, and public opinion tracking in dynamic online ecosystems.

7 Conclusion and future work

This paper presented an interpretable, topic-guided stance detection framework that integrates BERTopic for semantic topic discovery with RoBERTa for contextual stance classification, further enriched with sentiment cues to improve robustness in noisy, short-text environments such as Twitter. Through evaluations on two benchmark datasets such as SemEval 2016 Task 6 and the COVID-19 Stance Dataset, our hybrid approach demonstrated consistent performance improvements over traditional deep learning and transformer-based baselines. The results confirmed that combining unsupervised topical structure with supervised contextual embeddings not only boosts classification accuracy but also provides transparent, explainable predictions, a critical advancement for computational social science applications.

Beyond empirical performance, this work contributes a significant methodological and societal step forward. The proposed pipeline bridges a long-standing gap between accuracy and interpretability in stance detection, enabling human-in-the-loop workflows, more transparent social discourse monitoring, and data-driven policy support. Furthermore, it showcases how topic–stance alignment can illuminate the underlying narratives shaping public opinion, aiding journalists, policymakers, and researchers in responding to emerging societal challenges such as misinformation, polarization, and ideological echo chambers.

7.1 Implications, risks, and responsible deployment

While the framework offers substantial benefits, deploying stance detection models in real-world environments entails notable risks and ethical considerations. If used irresponsibly, these systems could be weaponized for large-scale opinion profiling, targeted disinformation, or political manipulation especially in sensitive contexts like elections,

protests, or public health campaigns. Additionally, latent biases within training data or topic clustering processes could amplify harmful stereotypes or distort minority perspectives, contributing to digital polarization.

To mitigate these risks, we recommend several strategies:

- Bias auditing and fairness evaluation: Regular assessments using diverse datasets to minimize demographic or ideological bias.
- Human-in-the-loop decision-making: Incorporating expert oversight during deployment, especially in sensitive policy or security contexts.
- Explainable interfaces: Developing visual dashboards and stance-topic rationale explanations to ensure transparency for end-users.
- Governance and accountability frameworks: Involving ethicists, legal experts, and civil society in model oversight to prevent misuse.

7.2 Future directions

While the current work makes significant strides in bridging interpretability and performance for stance detection, several promising directions remain open for future exploration:

- Human-Centered Interpretability and Trust Evaluation: Future studies should go beyond quantitative coherence metrics to include human-centered evaluations, such as user studies and expert annotation tasks. These will assess how well topic-informed explanations align with human judgment and how effectively they support decision-making in domains like policy analysis, journalism, and online moderation.
- Real-Time and Streaming Deployment: Extending the framework to handle continuous, streaming social media data is a crucial next step. This involves developing adaptive topic modeling mechanisms, incremental stance classifiers, and real-time dashboards capable of tracking evolving discourse patterns as events unfold.
- Cross-Lingual, Cross-Domain, and Multimodal Expansion: Future research should evaluate and adapt the model for multilingual environments using cross-lingual transformers (e.g., XLM-R, mBERT) and explore multimodal fusion (e.g., combining text with images, videos, or network metadata). This will enhance stance detection in global contexts and improve robustness in visually driven platforms like TikTok or Instagram.
- Ablation, Model Compression, and Lightweight Deployment: Future work can include comprehensive ablation studies to quantify the contribution of each component (topic modeling, sentiment integration, contextual embeddings). In parallel, adopting lighter models (e.g., DistilBERT, ALBERT) and applying techniques such as pruning or quantization will support scalable, resource-efficient deployment in real-world settings.
- Bias Mitigation and Societal Impact Analysis: A deeper exploration of how stance-topic patterns influence public opinion formation, policy debates, and misinformation diffusion is needed. Additionally, developing bias auditing frameworks and fairness-aware stance models will be vital to ensure responsible AI deployment, especially in politically sensitive or high-stakes contexts.

- **Policy and Governance Integration:** Future research should also investigate how stance detection systems can be incorporated into governance workflows—such as early warning systems for misinformation, civic engagement tools, or media literacy platforms—while maintaining ethical safeguards, transparency, and accountability.

7.3 Cost considerations and practical implications

Deploying a stance detection pipeline in production involves trade-offs between computational cost, interpretability, and governance. A scalable cloud-based implementation of the proposed system can operate with moderate infrastructure costs (approximately USD 2,000–5,000 per year) for continuous opinion monitoring on public social media platforms. However, several additional cost considerations may arise:

- **Bias Auditing and Fairness Checks:** Requires the collection of balanced, annotated datasets and regular audits to detect and mitigate bias.
- **Human-in-the-Loop Oversight:** Ongoing expert involvement for interpreting model outputs, refining topic clusters, and ensuring responsible decision-making.
- **Compliance and Governance:** Additional resources may be needed for privacy-preserving data handling, legal compliance, and ethical review processes.

Despite these costs, the societal and operational benefits significantly outweigh the investment. Improved misinformation detection, enhanced situational awareness for policymakers, and better understanding of online discourse dynamics contribute directly to healthier digital ecosystems and more informed decision-making.

Author contributions

R.M.: Conceptualization, Methodology, Formal Analysis, Software Development, Writing – Original Draft, Project Administration, Writing – Final Review & Supervision. K.S.: Data Curation, Validation, Visualization, Writing – Review & Editing. K.S.R.: Investigation, Resources, Supervision, Writing – Review & Editing. P.S.: Methodology, Writing – Review & Editing. G.S.: Evaluation Design, Writing – Review & Editing.

Funding

The authors have not disclosed any funding.

Data availability

Authors utilized publicly accessible anonymized data available at *SemEval-2016 Task 6 Dataset*: <https://www.saifmoham.org/WebPages/STANCEDataset.htm> *COVID-19 English Tweets Stance Dataset*: <https://github.com/kglandt/stance-detection-in-covid-19-tweets>

Declarations

Conflict of interest

The authors declare there is no conflict of interest

Ethical approval

This research did not involve human participants, animal experiments, or clinical trials. All datasets used (SemEval-2016 Task 6, COVID-19 Stance Dataset, and other publicly available benchmark corpora) are openly accessible and have been previously released with appropriate ethical clearance by their original creators.

Consent to participate

Not applicable. The study relies solely on publicly available, anonymized datasets and does not involve direct interaction with human participants.

Consent to publish

Not applicable. The manuscript does not contain any individual person's data in any form (including images, videos, or case details).

Received: 16 July 2025 / Accepted: 31 October 2025

Published online: 26 November 2025

References

1. Rita P, Antonio N, Afonso A. Social media discourse and voting decisions influence: sentiment analysis in tweets during an electoral period. *Soc Netw Anal Mining*. 2023. <https://doi.org/10.1007/s13278-023-01141-3>.
2. Kubin E, von Sikorski C. The role of (social) media in political polarization: a systematic review. *Ann Int Commun Assoc*. 2021;45(3):188–206. <https://doi.org/10.1080/23808985.2021.1976070>.
3. Muthusami R, Saritha K. An investigation of public perception of COVID-19 vaccination administration on Twitter using sentiment classification and topic modeling. *Humanit Soc Sci Stud*. 2022;11(1):1–9.
4. Muthusami R, Bharathi A, Saritha K. COVID-19 outbreak: tweet-based analysis and visualization towards the influence of coronavirus in the world. *Journal of GED Organization*. 2020;33(2):534–49.
5. Muthusami R, Bharathi A. Stance detection and mobile app recommendation discourse on tweets. *Comput Intell*. 2019;35(4):1043–60. <https://doi.org/10.1111/coin.12228>.
6. Mohammad SM, Sobhani P, Kiritchenko S. Stance and sentiment in tweets. *ACM Trans Internet Technol*. 2017;17(3):1–26. <https://doi.org/10.1145/3003433>.
7. Blei DM, Lafferty JD. A correlated topic model of science. *Ann Appl Stat*. 2007;1(1):17–35. <https://doi.org/10.1214/07-AOAS114>.
8. Blei DM. Probabilistic topic models. *Commun ACM*. 2012;55(4):77–84. <https://doi.org/10.1145/2133806.2133826>.
9. Roberts ME, Stewart BM, Tingley D, Lucas C, Leder-Luis J, Gadarian SK, et al. Structural topic models for open-ended survey responses. *Am J Polit Sci*. 2014;58(4):1064–82. <https://doi.org/10.1111/ajps.12103>.
10. Muthusami R, Saritha K. A novelty analysis about an impact of tweets and Twitter bios on topic quality discovery using topic modeling. *Indian Natl Acad Eng Ser B*. 2022;103(5):1431–41. <https://doi.org/10.1007/s40031-021-00642-0>.
11. Muthusami R, Saritha K. Global analysis of COVID-19 clinical related trials. *J Microbiol Infect Dis*. 2020;10(4):183–7. <https://doi.org/10.5799/jmid.852215>.
12. Devlin J, Chang MW, Lee K, Toutanova K BERT: pre-training of deep bidirectional transformers for language understanding. In: *Proceedings of NAACL-HLT 2019*, 4171–4186. (2019) <https://doi.org/10.48550/arXiv.1810.04805>.
13. Brown TB, Mann B, Ryder N, Subbiah M, Kaplan J, Dhariwal P, et al. Language models are few-shot learners. *Adv Neural Inf Process Syst*. 2020;33:1877–901. <https://doi.org/10.48550/arXiv.2005.14165>.
14. Muthusami R, Mani Kandan N, Saritha K, et al. Investigating topic modeling techniques through evaluation of topics discovered in short texts data across diverse domains. *Sci Rep*. 2024;14:12003. <https://doi.org/10.1038/s41598-024-73944-4>.
15. Meng X, Yan X, Zhang K, et al. The application of large language models in medicine: a scoping review. *iScience*. 2024;27(5):109713. <https://doi.org/10.1016/j.isci.2024.109713>.
16. Pang B, Lee L. Opinion mining and sentiment analysis. *Found Trends Inf Retr*. 2008;2(1–2):1–135. <https://doi.org/10.1561/1500000011>.
17. Yang X, et al. Sentiment analysis and its application to social media data. In: *Proceedings of the 60th annual meeting of the association for computational linguistics*. (2022) <https://doi.org/10.48550/arXiv.2205.03036>.
18. Alharbi A, Hai AA, Aljurbua R, Obradovic Z. AI-driven sentiment trend analysis: enhancing topic modeling interpretation with ChatGPT. In: *Maglogiannis I, Iliadis L, Macintyre J, Avlonitis M, Papaleonidas A, editors. Artificial intelligence applications and innovations. AIAI 2024. IFIP advances in information and communication technology*, vol. 712. Berlin: Springer; 2024.
19. Alturayef N, Luqman H, Ahmed M. A systematic review of machine learning techniques for stance detection and its applications. *Neural Comput Appl*. 2023;35:5113–44. <https://doi.org/10.1007/s00521-022-07723-6>.
20. Yang Z, Dai Z, Yang Y, Carbonell J, Salakhutdinov R, Le QV. XLNet: generalized autoregressive pretraining for language understanding. *Adv Neural Inf Process Syst*. 2019;32:5753–63. <https://doi.org/10.48550/arXiv.1906.08237>.
21. Müller M, Salathé M, Kummervold PE. COVID-Twitter-BERT: a natural language processing model to analyse COVID-19 content on Twitter. *Front Artif Intell*. 2023;6:1023281. <https://doi.org/10.3389/frai.2023.1023281>.
22. AlDayel A, Magdy W. Stance detection on social media: state of the art and trends. *Inf Process Manage*. 2021;58(4):102597. <https://doi.org/10.1016/j.ipm.2021.102597>.
23. Grootendorst M BERTopic: neural topic modeling with a class-based TF-IDF procedure. *arXiv Preprint*. (2022) <https://doi.org/10.48550/arXiv.2203.05299>.
24. Anand T, Pratik V A framework for fine grained sentiment analysis on code-mixed language for social media user behaviours. *ITU Kaleidoscope: innovation and digital transformation for a sustainable world (ITU K)*, 1–8. (2024) <https://doi.org/10.23919/ITUK59762.2024.10580525>.
25. Zhang B, Dai G, Niu F, Yin N, Fan X, Wang S, Huang H A survey of stance detection on social media: New directions and perspectives. *arXiv Preprint*. (2024) <https://doi.org/10.48550/arXiv.2409.15690>.
26. Xuanyuan M, Xiao L, Duan M. Sentiment classification algorithm based on multi-modal social media text information. *IEEE Access*. 2021;9:33410–8. <https://doi.org/10.1109/ACCESS.2021.3061505>.
27. Chandrasekaran G, Nguyen TN, Jude D. Multimodal sentimental analysis for social media applications: a comprehensive review. *WIREs Data Min Knowl Discovery*. 2021;11(6):e1415. <https://doi.org/10.1002/widm.1415>.
28. Mohammad S, Kiritchenko S, Sobhani P, Zhu X, Cherry C A dataset for detecting stance in tweets. In: *Proceedings of the tenth international conference on language resources and evaluation (LREC'16)* (pp. 3945–3952). (2016)
29. Glandt K, Khanal S, Li Y, Caragea D, Caragea C. Stance detection in COVID-19 tweets. In: *Proceedings of the 59th annual meeting of the association for computational linguistics and the 11th international joint conference on natural language processing (Volume 1: Long Papers)* (pp. 1596–1611). Association for computational linguistics. (2021) <https://doi.org/10.18653/v1/2021.acl-long.127>.
30. Liu Y, Ott M, Goyal N, Du J, Joshi M, Chen D, Stoyanov V RoBERTa: a robustly optimized BERT pretraining approach. *arXiv Preprint*. (2019) <https://doi.org/10.48550/arXiv.1907.11692>.
31. Müller M, Salathé M, Kummervold PE COVID-Twitter-BERT: a natural language processing model to analyse COVID-19 content on Twitter. *arXiv Preprint*. (2020) <https://doi.org/10.48550/arXiv.2005.07503>.
32. McInnes L, Healy J, Melville J UMAP: uniform manifold approximation and projection for dimension reduction. *arXiv Preprint*. (2020) <https://doi.org/10.48550/arXiv.1802.03426>.
33. Raman R, Pattnaik D, Hughes L, Nedungadi P. Unveiling the dynamics of AI applications: a review of reviews using scientometrics and BERTopic modeling. *J Innov Knowl*. 2024;9(3):100517. <https://doi.org/10.1016/j.jik.2024.100517>.

34. Wang Z, Chen J, Chen J, et al. Identifying interdisciplinary topics and their evolution based on BERTopic. *Scientometrics*. 2024;129(12):7359–84. <https://doi.org/10.1007/s11192-024-05107-6>.
35. Tang Z, Pan X, Gu Z. Analyzing public demands on China's online government inquiry platform: a BERTopic-based topic modeling study. *PLoS ONE*. 2024;19(2):e0296855. <https://doi.org/10.1371/journal.pone.0296855>.
36. Hutto C, Gilbert E. VADER: a parsimonious rule-based model for sentiment analysis of social media text. *Proc Int AAAI Conf Web Soc Media*. 2014;8(1):216–25.
37. Barbieri F, Camacho-Collados J, Neves L, Espinosa-Anke L. TweetEval: unified benchmark and comparative evaluation for tweet classification. *arXiv Preprint*. (2020) <https://doi.org/10.48550/arXiv.2010.12421>
38. Hardalov M, Arora A, Nakov P, Augenstein I. A survey on stance detection for mis- and disinformation identification. *arXiv Preprint*. (2021) <https://doi.org/10.48550/arXiv.2103.00242>
39. Ahmad PN, Shah AM, Lee K, Muhammad W. Misinformation detection on online social networks using pretrained language models. *Inf Process Manag*. 2026;63(1):104342.

Publisher's Note

Springer Nature remains neutral with regard to jurisdictional claims in published maps and institutional affiliations.